Two-Way Assistant: A Knowledge Distillation Object Detection Method for Remote Sensing Images

Xi Yang, *Senior Member, IEEE*, Sheng Zhang, and Weichao Yang

***Abstract*— Due to resource constraints on edge devices, lightweight detection models are gaining popularity in remote sensing. However, achieving efficient performance with these models is challenging compared with traditional detection models. Knowledge distillation (KD) is a promising solution to this issue. However, previous KD methods in remote sensing often suffer from background noise and fail to address feature disparities among detectors. To address the above issues, we introduce the two-way assistant (TWA) distillation method for remote sensing object detection. TWA comprises two crucial modules: the compression assistant module (CPAM) and the multiscale adaptive assistant module (MAAM). CPAM reduces the background information and category interference by compressing and redistributing teacher model features to the student model. MAAM enhances feature knowledge through multiscale fusion, addressing feature disparities. Through extensive experiments conducted on two distinct types of remote sensing datasets, optical learning, vision and remote sensing laboratory (LEVIR) and SAR SSDD, our TWA demonstrates favorable performance across both single-stage and two-stage detectors. Especially, it achieves a performance of 82.5% (optical LEVIR dataset) and 95.4% (SAR SSDD dataset) in the AP_{50} metric, superior to existing state-of-the-art methods.**

***Index Terms*— Knowledge distillation (KD), object detection, remote sensing images, two-way assistant (TWA).**

I. Introduction

With the rapid development of deep learning, large models have made significant progress in the field of remote sensing detection. However, large models mean complex structures, computationally expensive, and storage intensive. These factors create challenges for them in terms of terminal development. For this reason, lightweight detectors are being developed. However, the existing lightweight detectors are usually unable to achieve competitive performance compared with classic heavy detectors. They are faced with some issues, such as weak features' representation ability, high false alarm, and imprecise regression of bounding boxes. To solve these problems, researchers propose model compression techniques. Common model compression techniques include knowledge distillation (KD) [1], lightweight model architectures [2], pruning [3], [4], and quantization [5]. KD is characterized by its interpretability, flexibility, and adaptability. It compresses models by transferring the dark knowledge from the teacher model to the student model, improving the latter's performance. The main distillation methods include features layer distillation and regression distillation [1], [6], [7], such as attentional transfer (AT) [8] and double similarity distillation (DSD) [9], which achieve knowledge transfer by mimicking the attention and similarity of the teacher model. In addition, Heo et al. [10] help to assist in the training of learning models by separating the foreground from the background. Kim et al. [11] instead focused on enhancing distillation by emphasis on background removal. Zhixing et al. [12] suggest that important features should be selected to improve generalized detectability. Wang et al. [13] find that the use of positional differences allowed students to imitate the teacher's more fine-grained information.

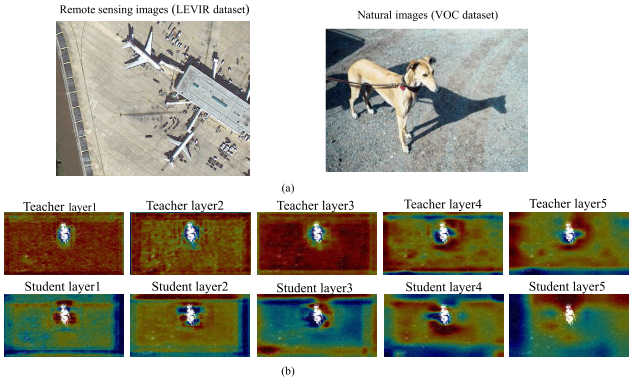


Fig. 1. Issues with KD methods in remote sensing detection. (a) Remote sensing images (from the LEVIR dataset) and natural images (from the VOC dataset). (b) Comparison of teacher–student feature layers.

However, the above methods do not fully consider remote sensing images multiscale targets and complex background characteristics. These characteristics bring the issues: 1) compared with natural images, remote sensing images with complex backgrounds usually cause interference with the target and pose problems associated with noise [14] [Fig. 1(a)]

Manuscript received 26 September 2023; revised 20 November 2023 and 2 January 2024; accepted 26 February 2024. Date of publication 29 February 2024; date of current version 8 March 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62372348 and in part by Shaanxi Outstanding Youth Science Fund under Project 2023-JC-JQ-53. *(Corresponding author: Weichao Yang.)*

Xi Yang is with the State Key Laboratory of Integrated Services Networks, School of Telecommunications Engineering, Xidian University, Xi'an 710071, China (e-mail: yangx@xidian.edu.cn).

Sheng Zhang is with Hangzhou Institute of Technology, Xidian University, Hangzhou 311231, China (e-mail: shengzhang@stu.xidian.edu.cn).

Weichao Yang is with Xi'an Institute of Space Radio Technology, Xi'an 710100, China (e-mail: yangwc@cast504.com).

Digital Object Identifier 10.1109/TGRS.2024.3371681

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:58:58 UTC from IEEE Xplore. Restrictions apply.

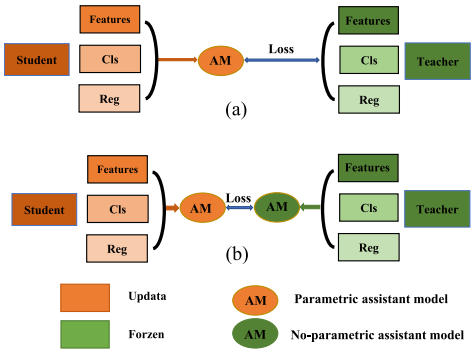


Fig. 2. Different KD frameworks for object detection. (a) In one-way assistant framework, the student model imitates the teacher's knowledge in a one-way manner through a teaching assistant module. (b) TWA framework by adding a parameter-free module to enable two-way distillation.

and 2) attention differences between teacher and student feature for remote sensing images [Fig. 1(b)] make the distillation process difficult. Usually, student models can access the wealth of dark knowledge from differences in teacher features. However, differences in noise and attention greatly affected the learning outcomes of the student detectors. In addition, traditional KD methods are overly dependent on the quality of the feature and regression results of the teacher's model, and their inaccuracy may directly affect the regression performance of the student's model, thus leading to the poor student model performance.

To address the problem of noise with small targets and complex backgrounds, some researchers propose the use of the mask method, which passes the features of the mask from the teacher model to the student model, called the one-way approach. However, the mask method can avoid some of the noise information. However, the features within the mask also contain noise. Therefore, an effective way to solve this problem is to improve the quality of teacher features. However, when distilling, the teacher network weights need to be frozen and cannot be selected using learnable weights. Inspired by this, we propose the compression assistant module (CPAM). The CPAM compresses and redistributes the high-dimensional feature information of the teacher model to obtain low-dimensional distribution information. By acting as a filter, the CPAM reduces background noise, focuses on the target feature information, and suppresses class interference.

For the problem of differences in teachers' and students' network features, some researchers propose weighting strategies. Introducing hyperparameters to balance the impact of large differences on learning. However, these methods do not consider the possibility that hyperparameters may weaken the capacity to characterize useful information. For this reason, the key to solving the problem is how to selectively balance the differences, rather than simply balancing all of them. Therefore, we propose the multiscale adaptive assistant module (MAAM). The MAAM includes a multiscale cross-focus module that combines different learnable parameters to fuse multiscale features. This not only enhances students' ability to learn detailed information from the teacher but also balances the differences between different features. To improve distillation by enhancing the learning of key information.

Briefly, this study proposes a two-way assistant (TWA) KD for remote sensing object detection. Unlike the one-way approach, TWA introduces an additional parameter-free module when processing teacher model features (Fig. 2). This two-way approach suppresses interference and optimizes the lightweight target detector by limiting teacher feature information. Experiments on different types of learning, vision and remote sensing laboratory (LEVIR) (optical) and SSDD (SAR) remote sensing datasets have demonstrated that the TWA method outperforms the existing [12], [13], [15], [16] KD method in terms of performance. In addition, the method is applicable to all types of detectors, including two-stage and one-stage detectors. Even the student model outperforms the teacher model on several metrics.

The contributions of this study are summarized as follows.

1) We propose a TWA distillation approach that incorporates a parameter-free assistant module and enables a two-way distillation approach. We validate the effectiveness of our method on multiple datasets.
2) We introduce a CPAM that suppresses spatial background noise and addresses attentional disparity issues by compressing and reallocating corresponding feature layers.
3) We present an MAAM that improves student access to detail and balances different feature representations of teachers through multiscale feature fusion.

II. Related Work

A. Object Detection

CNN-based detectors have significantly improved object detection in computer vision. These detection algorithms can be categorized into single-stage methods [17], [18], [19], [20] and two-stage methods [21], [22], [23]. Both approaches rely on large backbone networks to achieve high detection accuracy. However, the computational complexity of these methods poses a significant challenge for practical deployment, particularly on resource-limited devices, such as video surveillance, airborne, and satellite platforms. Traditional backbone networks like VGG and ResNet [24], due to their depth and large number of parameters, negatively impact the performance of detectors. Therefore, researchers have developed lightweight backbone, such as MobileNet [25], and detection models like SSD [18] and SSDLite [26].

Compared with natural images, remote sensing images have the characteristics of complex background and multiscale objects, and the above lightweight detectors usually do not achieve satisfactory detection results, necessitating the design of object detection algorithms for remote sensing data. Wang et al. [13] proposed spatial feature pyramids to fuse background information, allowing the model to focus more on small objects. Xu et al. [28] addressed spatial displacement problems by introducing a novel pseudoanchor module. In addition, Cheng et al. [29] addressed rotation-related issues by incorporating constraints and training the model to learn invariants. Cheng et al. [30] improved the classification of remote sensing scenes by incorporating a discriminative objective function-based approach into the model.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:58:58 UTC from IEEE Xplore. Restrictions apply.

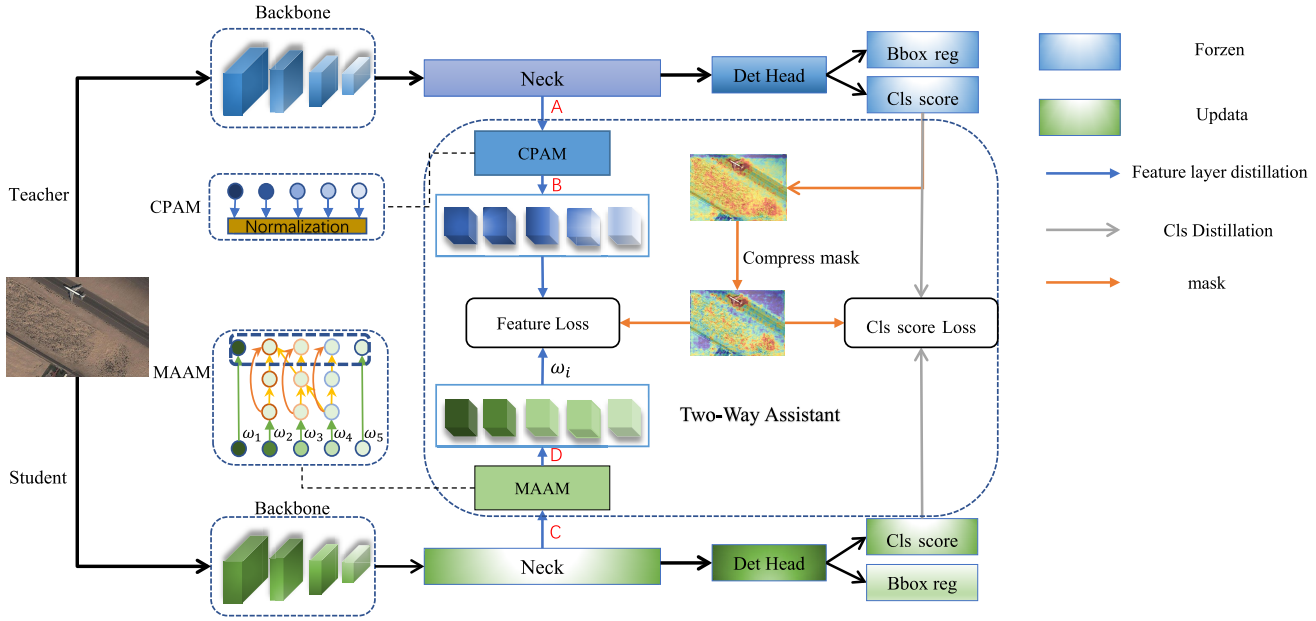


Fig. 3. Overall structure of TWA. The input images first enter the teacher's and student's backbone networks, respectively, for feature extraction, and then are fed into the classification and detection heads, respectively. We use the CPAM module to compress the extracted teacher's feature layer, which is then handed over to the student model for learning. The teacher's compressed knowledge is parsed through the MAAM to improve the student's ability to learn the details. Distillation of the FPN layers and classification heads is performed via the feature richness mask [12].

B. Knowledge Distillation

KD is a powerful technique for compressing models and improving the performance of smaller models. It was first introduced by Hinton et al. [1] for classification, enabling knowledge transfer from teacher models to smaller models through soft outputs. Currently, KD methods can be categorized into three types: logit-based [1], feature-based [13], [31], and relationship-based [32], [33], [34]. In logit-based and feature-based approaches, students acquire knowledge by directly or indirectly fitting the teacher's logit and certain features. On the other hand, relationship-based methods prioritize the exploration of relationships between different entities.

KD is initially used for classification tasks and achieves success [1], [8]. However, when KD is applied directly to complex tasks, such as object detection, its efficiency is hindered. Chen et al. [35] first attempt to apply KD to object detection. They use a method in which the student model directly emulates the features and regression results of the teacher model. Li et al. [36] further considers the issue of imbalance between foreground and background and propose a method that distinguishes positive and negative regions through region proposal network (RPN) features. Wang et al. [13] introduce a method that emulates fine-grained features by leveraging cross-position differences in feature responses. Zhang and Ma [16] propose a method based on attention-guided feature extraction. They simulate masks to extract critical foreground pixels. Therefore, Yang et al. [37] examine the influence of foreground and background on knowledge extraction, aiming to improve extraction performance by balancing feature differences. Yang et al. [38] introduced a KD method for remote sensing image detection, but it overlooked the investigation of relationships between different instances. They put forward a mask module to guide the student to mimic the core object features only. In our experiments, we observed that these mask-based one-way distillation strategies fail to achieve satisfactory performance, primarily due to the complexities associated with the data. Therefore, in our approach, based on two-way distillation strategies and starting from the teacher's feature knowledge, we enhance the student's ability to acquire crucial knowledge and distinguish between the background and foreground.

III. Methodology

The structure of our proposed TWA is shown in Fig. 3. The input images are fed into a backbone network of student and teacher models to extract their respective feature layers and then into the classification and detection heads, respectively. The teacher's feature layer is then fed into the CPAM to enhance the feature quality of the teacher model. Next, we incorporate the MAAM to assist in conveying the teacher's knowledge to the student model to facilitate learning. Finally, we select the richness score [12] of the compressed teacher features as the distillation criterion and implement the distillation technique to the feature pyramid network (FPN) layer and the classification head.

A. Compression Assistant Module

To address the issue of complex backgrounds in remote sensing images, traditional methods often employ mask modules to separate foreground and background information, enabling student models to learn key pixels from teacher models. However, these mask modules may also contain noise information. Moreover, due to the unique characteristics of remote sensing images, a standalone mask module may not effectively solve this problem. Therefore, we propose a parameter-free learning module called CPAM. Through the CPAM, we enhance the feature information of the teacher

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:58:58 UTC from IEEE Xplore. Restrictions apply.

model, which effectively eliminates background noise, aggregates essential feature information of the target, and suppresses interference information.

We obtain a feature map $f \in R^{C\times H\times W}$ from the backbone, where H, W, and C are the height, width, and number of channels, respectively, and f denotes the probability of an object. The probability y of the object belonging to class c given the feature f is expressed as $y = P(c \mid f)$.

To mitigate the impact of outliers on the data analysis and improve the accuracy and reliability of the results, we propose a compression and reconstruction method for f, excluding the maximum and minimum values. The reconstructed features are described as $f_l = \min f < f < \max f$, where $\min f$ and $\max f$ represent the minimum and maximum values across each channel in the feature at each layer, respectively.

Based on the obtained f_l, we perform compression to obtain f_c, as follows:

$$f_c = \frac{f_l}{\max(\|f_l\|_2, \epsilon)}. \tag{1}$$

Therefore, the probability of describing the feature as an object is obtained as follows:

$$y_c = P(c \mid f_c). \tag{2}$$

In addition, we utilize multiple dimensions and visualizations to demonstrate the functionality and effectiveness of CPAM comprehensively. A standardized approach is applied to ensure reliable and consistent comparative results and address the scattered dimensions by normalizing feature values as follows:

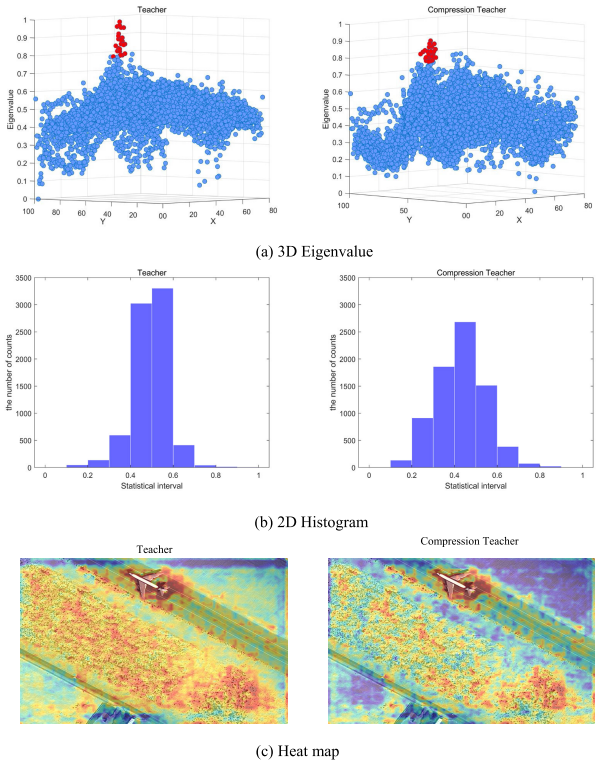
$$f_{\text{normalization}} = \frac{f - f_{\text{Min}}}{f_{\text{Max}} - f_{\text{Min}}}. \tag{3}$$

The feature map information is presented in a 3-D format through physical printing [Fig. 4(a)] to demonstrate the practical effectiveness of the CPAM, where red and blue represent the target and background locations, respectively. As shown in Fig. 4(b), 1-D statistical results of feature information before and after compression are displayed, with the feature information around the target and background locations becoming more compact and dispersed, respectively, after compression. The student model then focuses on learning the target location and disperses knowledge about the background, improving the feature quality of the teacher model, mitigating background interference, and enhancing the distillation effect.

Finally, to provide a more intuitive display of the results, we utilize visualizations for comparison, as shown in Fig. 4(c). From the image, it is evident that compared with before compression, the visualized feature heat map after compression exhibits a deeper intensity at the target location and a significantly lighter representation of background information. These improvements contribute to enhancing the quality of distillation.

B. Multiscale Adaptive Assistant Module

In the field of remote sensing object detection, different detectors often exhibit significant feature differences.



(a) 3D Eigenvalue

(b) 2D Histogram

(c) Heat map

Fig. 4. Visualization results before and after implementing CPAM. (a) Comparison chart for 3-D visualization. (b) Statistical comparison chart for characteristic value interval partitions. (c) Comparison of visualization feature maps.

To address this issue, traditional methods typically rely on introducing hyperparameters to balance these model differences. However, this balancing approach can compromise the teacher model's ability to effectively represent crucial feature information. Inspired by the residual structure from bidirectional feature pyramid network (BIFPN) [39] and the stacked form of multiscale spatial attention from multiscale adaptive attention module (MSAAM) [40], we propose MAAM (Fig. 5) for KD. Compared with BiFPN and MSAAM, we abandon complex structural forms to prevent the disruption of teacher features. Specifically, we discard the upsampling component of BiFPN and abandon the multibranch stacking of MSAAM to achieve a more streamlined module. MAAM enhances the student's understanding of detailed knowledge and enhances distillation performance.

We incorporate self-adaptive parameters into our proposed MAAM, which automatically selects objects of varying sizes for fusion and prioritizes features of different scales. Initially, we apply a convolution operation to the chosen student feature layer and introduce trainable parameters w_i to improve feature adaptability, resulting in s_i as follows:

$$s_i = \text{Conv}(w_i \cdot p_i). \tag{4}$$

Next, we obtain an intermediate layer g_i, computed as follows:

$$g_i = \text{Conv}\left(w^s_{i,1} \cdot s_i + w^s_{i,2} \cdot \text{Resize}(s_{i+1}); d_i\right). \tag{5}$$

Finally, we obtain the output t_i by concatenating the intermediate layer g_i with the resized feature map using

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:58:58 UTC from IEEE Xplore. Restrictions apply.

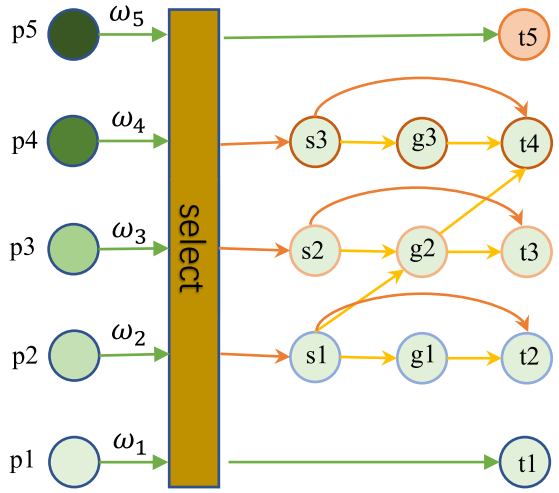


Fig. 5. Framework of the MAAM. p_i, s_i, g_i, w_i, and t_i represent the input layer, preprocessing layer, intermediate layer network, trainable training parameters, and output feature map, respectively.

the upsampling operation, where Resize denotes the resizing operation as follows:

$$t_i = \text{Conv}\left(w_{i,1}^s \cdot s_i + w_{i,1}^g \cdot g_i + w_{i,2}^g \cdot \text{Resize}(g_{i+1})\right). \quad (6)$$

We conduct a visualization experiment to facilitate the observation of the effects of the MAAM. We visualize the student model without distillation (point C in Fig. 3) and compare it with the student model with distillation (points C and D in Fig. 3). As shown in Fig. 6, in comparison with the nondistillation model, the distillation model exhibits a greater focus on the target's position and reduced noise. In addition, after distillation, points C and D exhibit feature maps focusing more on background knowledge and target information, respectively. Therefore, the MAAM can, therefore, help to improve distillation by filtering out background information and making the model more focused on detailed information.

C. Knowledge Distillation and Loss

We derive formulas of mask, FPN loss, class loss, and overall loss function for our proposed CPAM and MAAM.

1) Mask: y_c denotes the probability that the feature is an object. This conditional probability is utilized to construct a neural network model as follows:

$$y_c = P(c \mid f_c, \theta) \quad (7)$$

where the parameters are denoted by θ.

We obtain more comprehensive features by aggregating the probabilities across different categories using the following maximum function:

$$S = \max_c y_c = \max_c P(c \mid f_c, \theta). \quad (8)$$

We apply the richness of features from the teacher as follows:

$$\text{mask} = \max_{1 \le c' \le C} y_{C'}^t = \max_{1 \le c' \le C} P\left(c' \mid f_c^t, \theta^t\right) \quad (9)$$

where C represents the category of the object and $C > 1$.

2) FPN Loss: The FPN module is a crucial component of the object detection framework. It consists of high-resolution shallow and low-resolution deep layers for detecting small and large objects, respectively. Therefore, we apply a mask to the selected layers by calculating the FPN loss as follows:

$$L_{\text{FPN}} = \sum_{l=1}^{M} \frac{w_l}{N_l} \sum_{i=1}^{W} \sum_{j=1}^{H} \text{mask}_{lij} \sum_{c=1}^{C} \left(\phi_{\text{cpam}}\left(F_{lijc}^t\right) - \phi_{\text{maam}}\left(F_{lijc}^s\right)\right)^2 \quad (10)$$

where M denotes the selected number of FPN layers, l represents the index of the FPN layer, and i and j indicate the position of the corresponding feature map in terms of width (W) and height (H). Nt is calculated as $\sum_{i=1}^{W} \sum_{j=1}^{H}$, w_l represents learnable parameters that balance the influence of each layer on the loss. mask_{lij} represents the value at positions i and j of layer l. ϕ_{cpam} represents the CPAM operation applied to the teacher features. ϕ_{maam} represents the MAAM operation applied to the student feature layers.

3) Class Loss: The loss of the classification head is described as follows:

$$L_{\text{cls}} = \sum_{l=1}^{M} \frac{1}{N_l} \sum_{i=1}^{W} \sum_{j=1}^{H} \text{mask}_{lij} \sum_{c=1}^{C} \phi\left(y_{lijc}^t - y_{lijc}^s\right)^2 \quad (11)$$

where y_{lijc} represents the output of the student–teacher classification head and ϕ represents the binary cross-entropy function.

4) Overall Loss Function: The overall loss is a weighted sum of distillation loss and conventional detector loss as follows:

$$L = L_{\text{GT}} + \alpha L_{\text{FPN}} + \beta L_{\text{cls}} \quad (12)$$

where L_{GT} is the training loss of the student detector, and α and β are the hyperparameters used to balance different distillation losses.

IV. Experimental Results

We conduct experiments on various detection frameworks, backbones, and two datasets, namely, LEVIR and SSDD, to validate the effectiveness and robustness of our approach. This section presents the experimental details and provides an analysis of the results.

A. Experimental Setting

1) Datasets: We employ public remote object detection datasets (LEVIR and SSDD) including both optical and SAR types to ensure a comprehensive evaluation of the performance of the TWA. The datasets feature diverse scenes and objects with varying shapes and categories.

LEVIR dataset is prepared by Zou and Shi [41], this dataset comprises over 22 000 images. We use a random selection of 80% of the entire dataset as the training set and 20% of the entire dataset as the test set. The dataset includes three types of objects: airplanes, ships, and oil tankers.

SSDD dataset is prepared by Li et al. [42], and this dataset was the first publicly available SAR dataset for research.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:58:58 UTC from IEEE Xplore. Restrictions apply.

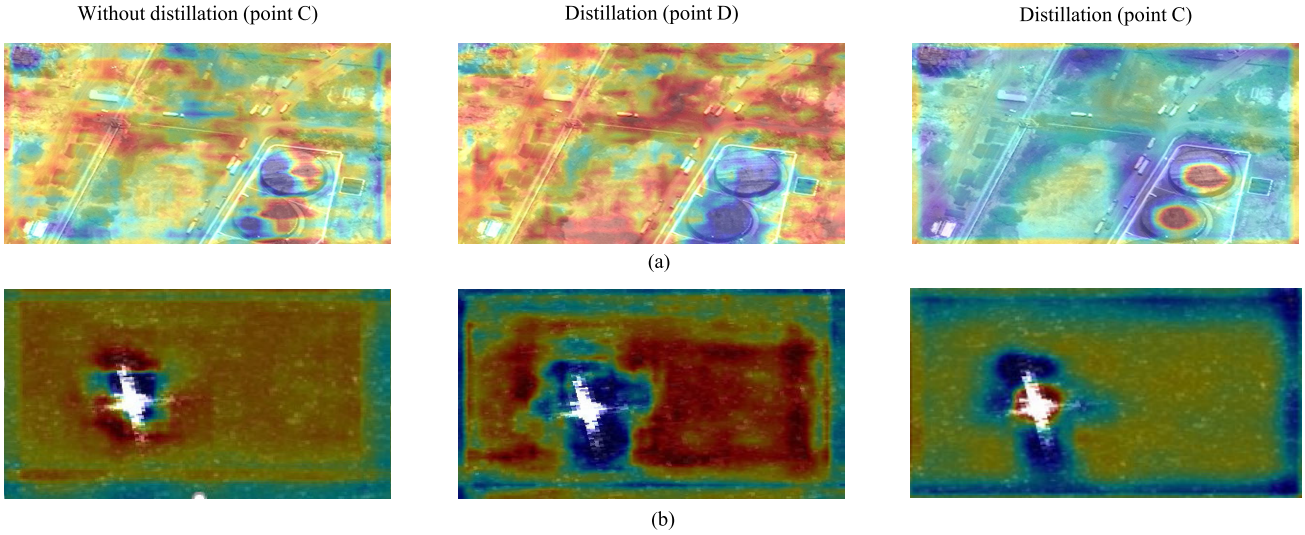


Fig. 6. Comparison of visualization results before and after implementing the MAAM on the LEVIR and SSDD images. (a) LEVIR images heat map. (b) SSDD images heat map.

It comprises 1160 images with about 500 × 500 pixel dimensions and includes 2456 instances of ships. Test images are identified with the suffixes "1" and "9" in their names, following the standard convention.

2) Implementation: Mean accuracy is our evaluation metric, comprising mAP, AP_{50}, AP_{75}, AP_S, AP_M, and AP_L. The latter three metrics measure performance for objects at different scales. Our implementation is based on mmdetection [44] with Pytorch, and the optimizer is configured as stochastic gradient descent (SGD) with a learning rate of 0.02, momentum of 0.9, and weight decay of 0.0001. We adopt a 2× learning rate schedule, where the initial learning rate is reduced by a factor of 10 between the 16th and 22nd epochs. In the experimental setup, all images are resized to 1000 × 600, except for the RetinaNet in the SSDD dataset, which is resized to 512 × 512.

B. Different Detection Frameworks

We validate the effectiveness of the proposed TWA across multiple detection frameworks, including anchor-based single-stage detectors (RetinaNet [20] and generalized focal loss (GFL) [45]) and two-stage detectors (Faster-region convolution neural networks (RCNN) [46]). In these experiments, the teacher models use ResNet50 or ResNet34 as the backbone, while the student model employs ResNet18 as the backbone. As shown in Table I, our approach consistently enhances the performance of student models, even surpassing teacher models on certain metrics.

On the LEVIR and SSDD datasets, Faster RCNN with ResNet18 outperforms its teacher model in terms of mAP, AP_{50}, and AP_{75}. In addition, GFL with ResNet18 also exhibits excellent performance. It is noteworthy that RetinaNet is more susceptible to the influence on the SSDD dataset, showing relatively poorer performance in its standalone with the help of the ResNet50 teacher detector. However, through the distillation method we propose, we successfully improve the model's performance across various metrics. These results demonstrate the adaptability and effectiveness of our approach on the SSDD and LEVIR remote sensing datasets.

C. Comparison With State-of-the-Art Methods

We compare our approach with existing KD methods, specifically including feature richness score (FRS) [12], focal and global distillation (FGD) [15], feature-based KD (FKD) [16], masked generative distillation (MGD) [43], adaptive reinforcement supervision distillation (ARSD) [37], and instance-aware distillation (InsDist) [31]. Among these, FKD and FGD are attention-guided distillation methods, emphasizing different parts of features based on learned attention weights. FRS utilizes the classification scores to highlight the importance of various features, enhancing the general detectability in the distillation process. MGD employs random mask operations to enhance the teacher model's stochastic expression ability, thereby improving the distillation's generalization capability. ARSD and InsDist are the two distillation methods specially designed for remote sensing object detection. To ensure experimental fairness, we conducted experiments on a two-stage detector (Faster-RCNN), where ResNet50 served as the backbone for the teacher detector, and ResNet18 served as the backbone for the student detector.

1) Results on LEVIR: The comparative results (Table II) show that our proposed method outperforms others in various metrics and even surpasses the performance of the teacher network in some cases. mAP and AP_{50} of the student model are 53.6% and 82.5%, respectively, higher than the teacher model's 52.9% and 82.1%. Compared with the baseline FRS, our method achieves significant improvements, with a 3% and 1.6% increase in mAP and AP_{50}, respectively.

2) Results on SSDD: A comparison of various methods on the SSDD dataset (Table II) demonstrates our method's superiority over the baseline FRS in various metrics. The AP_{75} improves by 10% over the baseline's 59.0%. Our method also outperforms existing alternative techniques. Furthermore, our approach enables the student model to outperform the teacher model in multiple metrics, showcasing the proposed approach's superiority with a smaller model.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:58:58 UTC from IEEE Xplore. Restrictions apply.

TABLE I
RESULTS OF THE PROPOSED METHOD WITH VARIOUS DETECTION FRAMEWORKS IN THE LEVIR AND SSDD DATASETS

Datase	LEVIR						SSDD					
Model	$mAP(\%)$	$AP_{50}(\%)$	$AP_{75}(\%)$	$AP_S(\%)$	$AP_M(\%)$	$AP_L(\%)$	$mAP(\%)$	$AP_{50}(\%)$	$AP_{75}(\%)$	$AP_S(\%)$	$AP_M(\%)$	$AP_L(\%)$
Faster-RCNN Res50(teacher)	52.9	82.1	59.3	43.4	57.6	51.9	58.4	95.2	65.4	53.8	66.2	64.4
Faster-RCNN Res18(student)	50.5	80.5	54.6	38.5	55.4	49.9	57.2	93.1	65.0	54.2	63.9	50.6
ours	53.6	82.5	57.8	42.6	58.1	53.9	59.0	95.4	69.3	54.7	66.5	64.0
gain	**+3.1**	**+2.0**	**+3.2**	**+4.1**	**+2.7**	**+4.0**	**+1.8**	**+2.3**	**+4.3**	**+0.5**	**+2.6**	**+13.4**
Faster-RCNN Res34(teacher)	50.6	80.9	56.1	41.4	54.6	49.2	58.4	94.5	67.1	54.9	64.8	60.9
Faster-RCNN Res18(student)	50.5	80.5	54.6	38.5	55.4	49.9	57.2	93.1	65.0	54.2	63.9	50.6
ours	51.7	81.8	56.5	42.0	56.7	50.6	59.6	94.8	67.9	55.7	66.3	61.0
gain	**+1.2**	**+1.3**	**+1.9**	**+1.5**	**+1.3**	**+0.7**	**+2.4**	**+1.7**	**+2.9**	**+1.5**	**+2.4**	**+10.4**
GFL Res50(teacher)	56.3	83.4	62.1	40.5	61.9	57.2	58.5	93.6	66.7	53.2	68.2	61.2
GFL Res18(student)	53.2	81.3	58.6	38.6	57.7	54.7	56.4	93.6	62.2	51.8	65.1	59.8
ours	55.1	82.6	60.3	38.8	60.4	56.0	58.3	94.3	67.1	53.3	67.6	60.2
gain	**+1.9**	**+1.3**	**+1.7**	**+0.2**	**+2.7**	**+1.3**	**+1.9**	**+0.7**	**+4.9**	**+1.5**	**+2.5**	**+0.4**
RetinaNet Res50(teacher)	55.4	80.9	61.4	35.8	62.3	57.5	48.1	85.5	49.2	51.5	60.0	47.7
RetinaNet Res18(student)	53.6	79.6	59.4	34.8	60.0	55.8	45.6	82.0	44.8	36.5	59.4	51.6
ours	54.9	81.1	60.0	36.6	62.0	55.8	51.7	88.0	58.3	46.5	61.8	54.2
gain	**+1.3**	**+1.5**	**+0.6**	**+1.8**	**+2.0**	**+0**	**+6.1**	**+6.0**	**+13.5**	**+10.0**	**+2.4**	**+2.6**

TABLE II
COMPARISON WITH SATE-OF-THE-ART METHODS USING DIFFERENT DETECTION FRAMEWORKS

Datase	LEVIR						SSDD					
Model	$mAP(\%)$	$AP_{50}(\%)$	$AP_{75}(\%)$	$AP_S(\%)$	$AP_M(\%)$	$AP_L(\%)$	$mAP(\%)$	$AP_{50}(\%)$	$AP_{75}(\%)$	$AP_S(\%)$	$AP_M(\%)$	$AP_L(\%)$
Faster-RCNN Res50(teacher)	52.9	82.1	**59.3**	43.4	57.6	51.9	58.4	95.2	65.4	53.8	66.2	**64.4**
Faster-RCNN Res18(student)	50.2	80.5	54.6	38.5	55.4	49.9	57.2	93.1	65.0	54.2	63.9	50.6
FRS [12]	50.6	80.9	55.3	40.4	56.0	48.7	53.7	91.8	59.0	49.1	62.5	49.5
FKD [16]	51.1	80.6	55.9	41.8	56.0	50.6	56.9	94.2	65.6	54.6	61.6	52.5
MGD [43]	52.5	81.1	57.0	40.7	56.7	52.2	58.2	93.3	66.5	54.5	65.2	58.0
FGD [15]	50.5	81.3	55.7	38.0	54.6	50.8	56.5	93.5	66.5	52.7	63.8	54.5
ARSD [37]	52.9	**82.8**	57.9	**45.1**	57.7	50.7	56.8	93.5	65.9	**55.5**	62.1	30.9
InsDist [31]	52.6	81.8	57.4	41.6	57.3	51.3	57.9	94.4	64.3	55.3	64.4	47.0
ours	**53.6**	82.5	57.8	42.6	**58.1**	**53.9**	**59.0**	**95.4**	**69.3**	54.7	**66.5**	64.0

3) Analysis on Both SSDD and LEVIR: The results show that our baseline FRS method achieves favorable results on the optical LEVIR dataset; however, it does not yield satisfactory results on the SSDD dataset, with its performance degrading as the training progresses. Similarly, other methods, such as FKD, do not significantly improve certain metrics. The MGD approach performs significantly better on the LEVIR dataset, while its performance on the SAR SSDD dataset is average. For the KD method designed for remote sensing image, ARSD demonstrates the excellent performance on the LEVIR dataset with its employed small target weighted strategy, where both AP_{50} and AP_S metrics surpass our approach. However, on the SSDD dataset, the performance of ARSD is relatively poor. In contrast, InsDist achieves strong performance by leveraging the relationships among different target instances in the SSDD dataset. Our method, on the other hand, performs well on both the LEVIR and SSDD datasets, with multiple metrics outperforming both ARSD and InsDist, verifying its greater robustness and generalization ability.

D. Ablation Study

1) CPAM and MAAM: We conduct ablation experiments to evaluate each module's impact on KD. We examine the effects of the CPAM and MAAM and its combination on the performance of the student model. The experimental results (Table III) indicate that each module improves the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:58:58 UTC from IEEE Xplore. Restrictions apply.

TABLE III
ABLATION STUDY FOR VARIOUS DISTILLATION MODULES ON THE LEVIR DATASET

FRS	✓	✓	✓	✓
CPAP		✓		✓
MAAM			✓	✓
$mAP(\%)$	80.9	81.5	81.5	**82.5**

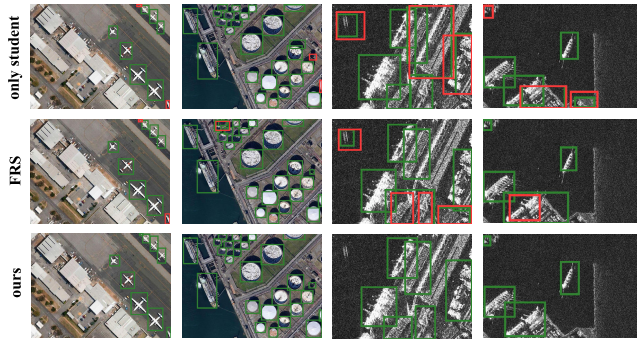


Fig. 7. Visual detection results of images from the LEVIR and SSDD datasets using Faster-RCNN with ResNet18. Green boxes indicate the results of model testing. Red boxes indicate objects that are not recognized or incorrectly recognized by the model. (a) LEVIR aircraft target image. (b) LEVIR oil barrel target image. (c) Image of a large ship for SSDD. (d) SSDD boat image.

performance of the student model. When CPAM and MAAM are incorporated into the baseline model, they enhance the performance of each model by 0.6%, while their combination achieves the best performance. These results indicate that each module positively impacts KD.

2) Layers of the MAAM: We conduct ablation experiments on MAAM to assess the impact of each layer on KD. As shown in Table IV, selecting features 1–3 for fusion results in the best overall performance. However, when additional high-level feature information is added, the overall performance slightly decreases. When choosing to use all high-level feature information, the overall performance is the worst. This is due to the presence of small targets in remote sensing images, while high-level feature information typically contains features related to larger targets. Therefore, when fusing high-level features with low-level features, the high-level features may have a detrimental impact on the lower level features, thus affecting the overall model performance.

E. Visualization Analysis

Under the same two-stage detector (Faster-RCNN), we utilize ResNet18 as the backbone for the student model and ResNet50 as the backbone for the teacher model. Various methods are employed, including an undistilled student detector, a detector distilled using FRS, and a detector distilled using the proposed TWA. Visual experiments on the SSDD and LEVIR datasets are conducted to assess the performance of the models.

1) Qualitative Analysis From Detection Results: As shown in Fig. 7, it is clearly visible that the TWA distillation method

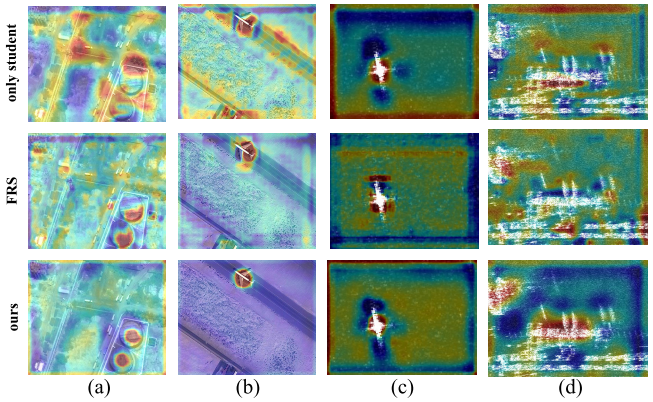


Fig. 8. Visual feature heat maps results of the baseline FRS approach and our proposed TWA method on the LEVIR and SSDD datasets. (a) LEVIR aircraft target thermal map. (b) LEVIR tank target heat map. (c) Single ship target heat map for SSDD. (d) SSDD ship target thermal map with complex background.

significantly improves the target detection performance of the student network. The undistilled student detector exhibits various types of false positives and misses on both the LEVIR and SSDD datasets. In Fig. 7(c), this detector generates multiple bounding boxes for the same target, and nonmaximum suppression (NMS) struggle to restrain these bounding boxes. In contrast, our method accurately localizes identical targets that are close to each other. Similarly, the detector distilled with FRS shows improvement in performance, but it also experiences instances of missed detections, especially in compact target scenarios, as shown in Fig. 7(b). Furthermore, in Fig. 7(d), when there is minimal difference between object categories, the FRS-distilled detector may struggle to correctly identify categories, while our method avoids these errors, highlighting the superior category discrimination capability of TWA.

2) Qualitative Analysis From Feature Heat Maps: As depicted in Fig. 8, we gain a clear understanding of the differences between different detectors through the use of heatmaps. First, for the undistilled student detector, the feature weights of other regions show higher values than the target region. It also shows that the complex background of remote sensing images has a relatively large influence on the performance of the detector. In addition, although the student model distilled with FRS shows improvement to some extent, its performance is not as robust as TWA. We can distinctly observe that at the edges of objects, the corresponding weights are relatively smaller, while in regions with rich feature information, such as the central area of the target, the feature weights are relatively higher. Simultaneously, the background is filtered out by CPAM and MAAM, leading to superior performance in our student detector.

The aforementioned conclusions are validated on the SSDD and LEVIR datasets. In the future, we plan to further validate these findings using additional datasets.

F. Analysis of Undesirable Results and Future Work

When dealing with specific types of noise, the TWA method may not perform optimally, as illustrated in Fig. 9. In the left representation, red indicates the prediction results, while

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:58:58 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ABLATION STUDY OF THE MAAM ON THE LEVIR DATASET

Layer1	✓	✓	✓	✓				✓	✓	✓
Layer2	✓	✓			✓	✓		✓	✓	✓
Layer3			✓	✓	✓	✓	✓	✓	✓	✓
Layer4	✓		✓		✓		✓		✓	✓
Layer5		✓		✓		✓	✓			✓
$mAP(\%)$	82.0	81.7	81.6	81.4	81.6	81.3	81.2	**82.5**	82.4	82.1

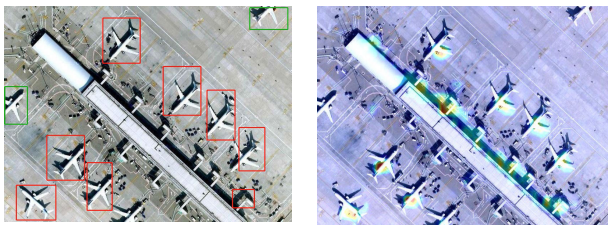


Fig. 9. Unsatisfactory detection results in the LEVIR dataset, red indicates the prediction results, while green represents the instances of missed detections.

green represents the instances of missed detections. In the right representation, we provide a visualization of feature heatmaps. It is evident that the TWA method tends to focus more on the central part of the target, making it susceptible to missed detections caused by edge noise. Second, due to the use of the MAAM, as seen in the heatmaps, the detector appears to excessively emphasize fine-grained details. This can potentially lead to overfitting, particularly when encountering background interference that is similar to the target.

V. CONCLUSION

This study proposes an effective distillation method for object detection in remote sensing. Our novel approach uses a two-way distillation method, incorporating a parameter-free model to preprocess teacher knowledge. In addition, we introduce two key modules: CPAM and MAAM. The CPAM compresses the teacher's feature knowledge, filters harmful features, and enhances useful feature information. The MAAM enhances the learning capability of the model through feature fusion, improving the distillation effect. The high performance of our approach demonstrates that it can be applied to modern detection frameworks at all stages.

However, the TWA method may underperform with certain types of noise and overemphasize fine details. Therefore, in future work, we aim to improve noise handling, optimize robustness, and explore attention modules. And, TWA has great potential for image tasks, advancing model compression and improving the efficiency of deep learning.

REFERENCES

[1] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.

[2] K. Han et al., "GhostNets on heterogeneous devices via cheap operations," *Int. J. Comput. Vis.*, vol. 130, no. 4, pp. 1050–1069, Apr. 2022.

[3] S. Han, J. Pool, J. Tran, and W. Dally, "Learning both weights and connections for efficient neural network," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 1135–1143.

[4] X. Yang, J. Zhang, C. Chen, and D. Yang, "An efficient and lightweight CNN model with soft quantification for ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5230713.

[5] R. Gong et al., "Differentiable soft quantization: Bridging full-precision and low-bit neural networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4852–4861.

[6] R. Zhang, S. Tang, L. Liu, Y. Zhang, J. Li, and S. Yan, "High resolution feature recovering for accelerating urban scene parsing," *Int. J. Comput. Appl.*, vol. 18, no. 1, pp. 1156–1162, Feb. 2018.

[7] S. I. Mirzadeh, M. Farajtabar, A. Li, N. Levine, A. Matsukawa, and H. Ghasemzadeh, "Improved knowledge distillation via teacher assistant," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 4, pp. 5191–5198.

[8] S. Zagoruyko and N. Komodakis, "Paying more attention to attention: Improving the performance of convolutional neural networks via attention transfer," 2016, *arXiv:1612.03928*.

[9] Y. Feng, X. Sun, W. Diao, J. Li, and X. Gao, "Double similarity distillation for semantic image segmentation," *IEEE Trans. Image Process.*, vol. 30, pp. 5363–5376, 2021.

[10] B. Heo, M. Lee, S. Yun, and J. Y. Choi, "Knowledge transfer via distillation of activation boundaries formed by hidden neurons," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2019, vol. 33, no. 1, pp. 3779–3787.

[11] J. Kim, S. Park, and N. Kwak, "Paraphrasing complex network: Network compression via factor transfer," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 2760–2769.

[12] D. Zhixing et al., "Distilling object detectors with feature richness," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 5213–5224.

[13] T. Wang, L. Yuan, X. Zhang, and J. Feng, "Distilling object detectors with fine-grained feature imitation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 4933–4942.

[14] G. Cheng, G. Wang, and J. Han, "ISNet: Towards improving separability for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5623811.

[15] Z. Yang et al., "Focal and global knowledge distillation for detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4643–4652.

[16] L. Zhang and K. Ma, "Improve object detection with feature-based knowledge distillation: Towards accurate and efficient detectors," in *Proc. Int. Conf. Learn. Represent.*, 2021, pp. 1–14.

[17] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[18] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.*, Oct. 2016, pp. 21–37.

[19] J. Redmon and A. Farhadi, "YOLO9000: Better, faster, stronger," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 7263–7271.

[20] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 2980–2988.

[21] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2961–2969.

[22] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2014, pp. 580–587.

[23] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:58:58 UTC from IEEE Xplore. Restrictions apply.

[24] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.
[25] A. G. Howard et al., "MobileNets: Efficient convolutional neural networks for mobile vision applications," 2017, *arXiv:1704.04861*.
[26] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520.
[27] X. Sun, Y. Liu, Z. Yan, P. Wang, W. Diao, and K. Fu, "SRAF-Net: Shape robust anchor-free network for garbage dumps in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6154–6168, Jul. 2021.
[28] T. Xu, X. Sun, W. Diao, L. Zhao, K. Fu, and H. Wang, "ASSD: Feature aligned single-shot detection for multiscale objects in aerial imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607117.
[29] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.
[30] G. Cheng, C. Yang, X. Yao, L. Guo, and J. Han, "When deep learning meets metric learning: Remote sensing image scene classification via learning discriminative CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2811–2821, May 2018.
[31] C. Li, G. Cheng, G. Wang, P. Zhou, and J. Han, "Instance-aware distillation for efficient object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5602011.
[32] J. Zhu et al., "Complementary relation contrastive distillation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jul. 2021, pp. 9260–9269.
[33] F. Tung and G. Mori, "Similarity-preserving knowledge distillation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1365–1374.
[34] Y. Chen, M. Lin, Z. He, K. Polat, A. Alhudhaif, and F. Alenezi, "Consistency- and dependence-guided knowledge distillation for object detection in remote sensing images," *Expert Syst. Appl.*, vol. 229, Nov. 2023, Art. no. 120519.
[35] G. Chen, W. Choi, X. Yu, T. Han, and M. Chandraker, "Learning efficient object detection models with knowledge distillation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 742–751.
[36] Q. Li, S. Jin, and J. Yan, "Mimicking very efficient network for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6356–6364.
[37] Y. Yang et al., "Adaptive knowledge distillation for lightweight remote sensing object detectors optimizing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5623715.
[38] Y. Yang, X. Sun, W. Diao, D. Yin, Z. Yang, and X. Li, "Statistical sample selection and multivariate knowledge mining for lightweight detectors in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5626414.
[39] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10781–10790.
[40] C. Wang, Z. Zhao, F. Wang, and Q. Li, "MSAAM: A multiscale adaptive attention module for IoT malware detection and family classification," *Secur. Commun. Netw.*, vol. 2022, pp. 1–14, Jun. 2022.
[41] Z. Zou and Z. Shi, "Random access memories: A new paradigm for target detection in high resolution aerial remote sensing images," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1100–1111, Mar. 2018.
[42] J. Li, C. Qu, and J. Shao, "Ship detection in SAR images based on an improved faster R-CNN," in *Proc. SAR Big Data Era, Models, Methods Appl.*, 2017, pp. 1–6.
[43] Z. Yang, Z. Li, M. Shao, D. Shi, Z. Yuan, and C. Yuan, "Masked generative distillation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2022, pp. 53–69.
[44] K. Chen et al., "MMDetection: Open MMLab detection toolbox and benchmark," 2019, *arXiv:1906.07155*.
[45] X. Li et al., "Generalized focal loss: Learning qualified and distributed bounding boxes for dense object detection," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 21002–21012.
[46] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 1137–1149.



Xi Yang (Senior Member, IEEE) received the B.Eng. degree in electronic information engineering and the Ph.D. degree in pattern recognition and intelligence system from Xidian University, Xi'an, China, in 2010 and 2015, respectively.

From 2013 to 2014, she was a Visiting Ph.D. Student with the Department of Computer Science, University of Texas at San Antonio, San Antonio, TX, USA. In 2015, she joined the State Key Laboratory of Integrated Services Networks, School of Telecommunications Engineering, Xidian University, where she is currently a Professor in communications and information systems. She has authored or coauthored over 60 articles in refereed journals and proceedings, including IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS (TNNLS), IEEE TRANSACTIONS ON CYBERNETICS (TCYB), IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), Computer Vision and Pattern Recognition (CVPR), International Conference on Computer Vision (ICCV), and Association for Computing Machinery International Conference on Multimedia (ACM MM). Her research interests include image/video processing, computer vision, and machine learning.



Sheng Zhang received the B.Eng. degree in communication engineering from Wuhan University of Science and Technology, Wuhan, Hubei, China, in 2022. He is currently pursuing the M.S. degree with the Hangzhou Institute of Technology, Xidian University, Hangzhou, China.

His research interests include deep learning and lightweight object detection.



Weichao Yang received the Ph.D. degree in signal and information processing from Harbin Engineering University, Harbin, China, in 2012.

He is currently with Xi'an Institute of Space Radio Technology, Xi'an, China. His research interests include image processing and spectrum sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:58:58 UTC from IEEE Xplore. Restrictions apply.